ANNAMALAI UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT TITLE

PREDICTING LEVELS OF DAMAGES TO BUILDINGS CAUSED BY EARTHQUACKE

Project Review - 1

Submitted by:

B.E - CSE(DATASCIENCE) FINAL YEAR

1. ARULSELVAM C - 2136110002
2. GOWTHAM R - 2136110047
3. PRANAVVARSHAN A T - 2136110039
4. RAJADURAI P - 2136110042

SIGNATURE OF GUIDE

(Dr. R. Ragupathy)

Abstract

The 2015 Nepal earthquake, a catastrophic 7.8 magnitude event, caused severe infrastructure damage and a tragic loss of life. This disaster underscored the urgent need for improved disaster management and preparedness. Predicting building damage levels post-earthquake is critical for effective response and resource allocation during such crises. This project utilizes deep learning models to predict damage levels to buildings using features such as structural details, building materials, and geographical information. By analyzing data from the 2015 Nepal earthquake, the project aims to understand patterns in building vulnerability. The insights generated will help authorities prioritize rescue operations and allocate resources efficiently. Additionally, the findings can guide urban planning to minimize risks in earthquake-prone areas. This works aims to enhance disaster resilience and preparedness, ensuring safer communities and improving disaster response strategies for future seismic events.

Introduction

The 2015 Nepal Earthquake: A Catastrophic Event

On April 25, 2015, Nepal experienced a catastrophic earthquake measuring 7.8 on the Richter scale, centered near Barpak in the Gorkha District. This seismic event, one of the worst in Nepal's history, caused widespread devastation across the country, affecting urban centers such as Kathmandu and remote rural areas. Over 9,000 lives were lost, more than 22,000 individuals sustained injuries, and millions were left homeless. The economic impact was equally staggering, with damages estimated at around $10 billion, nearly half of Nepal's GDP at the time. Many buildings, especially those not designed to withstand seismic forces, collapsed completely, exacerbating the human and economic toll.

Challenges in Post-Earthquake Damage Assessment

Traditional methods for assessing building damages, such as manual inspections, are resource-intensive and time-consuming. These methods often struggle to provide rapid and consistent results during large-scale disasters. In regions like Nepal, where infrastructure is diverse and often inadequately documented, the challenges become even more pronounced. The urgent need for innovative solutions that can quickly and accurately predict building damage levels is evident. By enabling faster response times, such solutions can save lives and optimize the allocation of limited resources during emergencies.

Advances in Machine Learning for Disaster Management

In recent years, advancements in machine learning (ML) and deep learning (DL) have shown immense potential in automating damage assessment. These technologies utilize historical data, structural characteristics, and environmental factors to predict outcomes with high accuracy. For example, convolutional neural networks (CNNs) have been successfully employed to analyze aerial imagery of damaged areas, while recurrent neural networks (RNNs) have been used to interpret sequential seismic data. By integrating such techniques, predictive models can provide actionable insights, bridging the gap between data collection and decision-making.

This project focuses on developing a deep learning-based model for predicting building damage levels using data from the 2015 Nepal earthquake. By incorporating features such as structural materials, construction quality, and geographical attributes, the model aims to deliver accurate and reliable predictions. The outcomes of this research will contribute to disaster preparedness, risk mitigation, and urban planning, ultimately reducing the human and economic impacts of future earthquakes.

Objectives of the Project

- Develop a deep learning-based predictive model to estimate building damage levels caused by earthquakes.
- Analyze the key factors influencing building damages, including structural materials, construction quality, and geographical attributes.
- Leverage data from the 2015 Nepal earthquake to validate and enhance the model's accuracy.
- Facilitate rapid disaster response and resource allocation through real-time damage predictions.
- Contribute to safer urban planning by identifying high-risk building types and areas.

Application Areas of the Project

1. **Disaster Management**: Enabling quick damage assessments and efficient response planning.
2. **Urban Planning**: Guiding the development of earthquake-resilient infrastructure.
3. **Insurance Industry**: Assisting in accurate risk assessment and claim estimations.
4. **Research and Policy Making**: Providing a data-driven basis for seismic risk mitigation strategies.

Literature Review

Journal	Authors	Title	Year	Key Findings	Abstract	Dataset	Technique Used
International Journal of Disaster Risk Reduction	Bai, Y., Adriano, B., Mas, E., & Koshimura, S.	Building Damage Assessment in the 2015 Gorkha, Nepal Earthquake Using Only Post-Event Dual Polarization Synthetic Aperture Radar Imagery	2017	Challenges in distinguishing between damaged and undamaged urban areas using single polarimetric features	Explores the use of post-event dual polarimetric synthetic aperture radar images for earthquake damage assessment, focusing on the 2015 Nepal earthquake.	Post-event SAR images	Dual Polarization Synthetic Aperture Radar
IEEE Geoscience and Remote Sensing Letters	Mangalraj, P., et al.	Post-earthquake Building Damage Detection Using Deep Learning	2021	Effective identification of damaged structures using imagery data	Presents a deep learning approach for detecting building damage after earthquakes, demonstrating the model's effectiveness.	Post-earthquake imagery	Deep Neural Networks
Earthquake Engineering & Structural Dynamics	Gautam, D., & Rodrigues, H.	Damage Scenario of Reinforced Concrete Buildings in the 2015 Nepal Earthquakes	2015	Analysis of causes and types of damage to reinforced concrete buildings	Summarizes the damage to reinforced concrete buildings during the 2015 Nepal earthquakes.	Field survey data	Structural Analysis
ISPRS Journal of Photogrammetry and	Ghorbanzadeh, O., et al.	Evaluation of Deep Learning Models for Building Damage	2020	Insights into deep learning models in real emergency	Illustrates the use of earth observation and deep learning	Satellite imagery	Convolutional Neural Networks (CNN)

Remote Sensing		Mapping in Emergency Response Settings		response scenarios	technologies in predicting building damage.		
Remote Sensing Letters	Wang, J., et al.	Superpixel-Based Building Damage Detection from Post-earthquake Very High Resolution Imagery	2019	Combines deep neural networks with superpixel segmentation for damage detection	Introduces a novel approach for detecting building damage from high-resolution imagery, applied to the 2015 Nepal earthquake.	High-resolution post-earthquake imagery	Deep Neural Networks with Superpixel Segmentation
Structural Safety	Demertzis, K., et al.	A Comparative Evaluation of Machine Learning Algorithms for the Prediction of R/C Buildings' Seismic Damage	2022	Effectiveness of various machine learning algorithms in seismic damage prediction	Evaluates various machine learning algorithms to predict seismic damage in reinforced concrete buildings.	Simulated earthquake damage dataset	Machine Learning Algorithms (Random Forest, SVM, etc.)
Journal of Seismology	Kostinakis, K., et al.	Classification of Buildings' Potential for Seismic Damage by Means of Artificial Intelligence Techniques	2021	Artificial intelligence techniques for pre-seismic vulnerability assessments	Explores the use of artificial intelligence techniques to classify buildings based on their potential for seismic damage.	Regional building vulnerability dataset	Artificial Neural Networks (ANN)
Journal of Earthquake Engineering	Gautam, D., & Chaulagain, H.	Seismic Performance and Post-Earthquake Reconstruction of Stone in Mud	2015	Highlights issues and strategies for rebuilding traditional masonry buildings	Discusses the seismic performance and reconstruction of traditional	Post-earthquake field survey data	Structural Performance Analysis

		Mortar Masonry Buildings in Nepal			masonry buildings in Nepal following the 2015 earthquake sequence.		
Natural Hazards	Zhu, Z., et al.	Earthquake Building Damage Detection Based on Synthetic-Aperture Radar: A Case Study of the 2015 Nepal Earthquake	2022	Framework for semi-automated building damage assessment using remote-sensing data	Presents a framework for building damage assessment due to earthquakes using remote-sensing data, applied to the 2015 Nepal earthquake.	SAR data	Semi-Automated Analysis Using Synthetic Aperture Radar
Remote Sensing	Li, Z., et al.	Rapid Assessment of Building Damage Using Multi-Source Data: A Case Study of the 2015 Nepal Earthquake	2019	Rapid building damage assessment through multi-source data	Discusses a method for rapid building damage assessment using multi-source data, with a focus on the 2015 Nepal earthquake.	Multi-source post-earthquake datasets	Data Fusion and Statistical Analysis
Remote Sensing Letters	Zhu, B., et al.	Deep Learning for Earthquake Disaster Assessment: Objects, Data, Models, and Challenges	2023	Comprehensive analysis of deep learning applications and challenges in disaster assessment	Analyzes numerous articles to investigate the status quo, development, and challenges of deep learning for earthquake disaster assessment.	Disaster assessment datasets	Convolutional Neural Networks (CNN)

Journal of Infrastructure Preservation and Resilience	Zhang, Y., et al.	Strong Ground Motion Data of the 2015 Gorkha Nepal Earthquake	2015	Provides strong ground motion data for seismic hazard mapping	Offers ground motion data from the 2015 Nepal earthquake, useful for seismic hazard mapping and building damage prediction.	Strong motion sensor data	Data Analysis and Modeling

References

1. Bai, Y., Adriano, B., Mas, E., & Koshimura, S. (2017). Building Damage Assessment in the 2015 Gorkha, Nepal Earthquake Using Only Post-Event Dual Polarization Synthetic Aperture Radar Imagery. *International Journal of Disaster Risk Reduction*.

2. Mangalraj, P., et al. (2021). Post-earthquake Building Damage Detection Using Deep Learning. *IEEE Geoscience and Remote Sensing Letters*.

3. Gautam, D., & Rodrigues, H. (2015). Damage Scenario of Reinforced Concrete Buildings in the 2015 Nepal Earthquakes. *Earthquake Engineering & Structural Dynamics*.

4. Ghorbanzadeh, O., et al. (2020). Evaluation of Deep Learning Models for Building Damage Mapping in Emergency Response Settings. *ISPRS Journal of Photogrammetry and Remote Sensing*.

5. Wang, J., et al. (2019). Superpixel-Based Building Damage Detection from Post-earthquake Very High Resolution Imagery. *Remote Sensing Letters*.

6. Demertzis, K., et al. (2022). A Comparative Evaluation of Machine Learning Algorithms for the Prediction of R/C Buildings' Seismic Damage. *Structural Safety*.

7. Kostinakis, K., et al. (2021). Classification of Buildings' Potential for Seismic Damage by Means of Artificial Intelligence Techniques. *Journal of Seismology*.

8. Gautam, D., & Chaulagain, H. (2015). Seismic Performance and Post-Earthquake Reconstruction of Stone in Mud Mortar Masonry Buildings in Nepal. *Journal of Earthquake Engineering*.

9. Zhu, Z., et al. (2022). Earthquake Building Damage Detection Based on Synthetic-Aperture Radar: A Case Study of the 2015 Nepal Earthquake. *Natural Hazards*.

10. Li, Z., et al. (2019). Rapid Assessment of Building Damage Using Multi-Source Data: A Case Study of the 2015 Nepal Earthquake. *Remote Sensing*.

11. Zhu, B., et al. (2023). Deep Learning for Earthquake Disaster Assessment: Objects, Data, Models, and Challenges. *Remote Sensing Letters*.

12. Zhang, Y., et al. (2015). Strong Ground Motion Data of the 2015 Gorkha Nepal Earthquake. *Journal of Infrastructure Preservation and Resilience*.